

Mutual Fund Analytics

Capstone Project Report

Submitted By:

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Internship Company:

Bluestock Fintech

Technologies Used:

Python · Pandas & NumPy · SQLite · SQLAlchemy
Matplotlib & Plotly · Jupyter Notebook · Power BI · Git & GitHub

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Abstract

This project, **Mutual Fund Analytics**, was developed to analyze and visualize mutual fund data using Python, SQLite, and Power BI. The project started with collecting and cleaning multiple datasets related to mutual funds, investor transactions, portfolio holdings, SIP inflows, and benchmark indices. After cleaning the data, a SQLite database was created to store the processed information in a structured format.

Exploratory Data Analysis (EDA) was performed to identify trends in NAV, AUM, SIP inflows, investor demographics, and category-wise investments through various charts and visualizations. Performance analytics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), and Conditional Value at Risk (CVaR) were calculated to evaluate fund performance and risk. Advanced analytics, including investor cohort analysis, SIP continuity analysis, sector concentration (HHI), and a simple fund recommendation system based on risk appetite, were also implemented.

Finally, an interactive four-page Power BI dashboard was developed to present key insights related to industry performance, fund performance, investor analytics, and market trends. The project demonstrates how data engineering, financial analytics, and business intelligence techniques can be combined to support better investment analysis and decision-making.

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1 Introduction

The mutual fund industry has become one of the most popular investment options for individuals because it provides professional fund management, portfolio diversification, and long-term wealth creation opportunities. As the number of mutual fund schemes and investors continues to grow, analyzing large volumes of financial data has become essential for understanding fund performance, investor behavior, and market trends. Data analytics helps transform raw financial information into meaningful insights that support informed investment decisions.

This project, **Mutual Fund Analytics Dashboard**, focuses on building a complete analytics solution using Python, SQLite, and Power BI. The architecture spans across three core domains:

- **Data Engineering:** Collecting, cleaning, and transforming multiple mutual fund datasets, followed by storing the processed data in a structured SQLite database.
- **Financial Analytics:** Applying exploratory and quantitative techniques to evaluate fund performance, measure investment risk, and uncover hidden trends.
- **Business Intelligence:** Building an interactive dashboard to present complex metrics in a visual, decision-ready format.

The project leverages **Exploratory Data Analysis (EDA)** to track critical shifts in Net Asset Value (NAV), Assets Under Management (AUM), SIP inflows, investor demographics, and category-wise fund performance. To rigorously assess the risk-return profile of different schemes, advanced financial metrics are calculated, including:

CAGR · Sharpe Ratio · Sortino Ratio · Alpha · Beta · Maximum Drawdown · Value at Risk (VaR) · Conditional Value at Risk (CVaR)

An interactive four-page **Power BI dashboard** has been developed to present the results in a simple and user-friendly manner. The dashboard allows users to seamlessly explore industry trends, compare fund performance, analyze investor behavior, and study market movements through dynamic charts, filters, and drill-through features.

Overall, this project demonstrates how data engineering, financial analytics, and business intelligence tools can be combined to convert raw mutual fund data into valuable business insights. The developed solution can help investors, financial analysts, and fund management companies better understand market performance and make more informed investment decisions.

2 Project Objectives

The main objective of this project is to develop a complete mutual fund analytics solution that processes raw financial data and converts it into meaningful insights through data analysis and visualization. The project aims to support better investment analysis by combining data engineering, financial analytics, and business intelligence techniques.

The specific objectives of this project are:

- **Data Collection & ETL:** Collect and clean mutual fund datasets from different sources.
- **Structured Storage:** Store the cleaned data in a structured SQLite database.
- **Exploratory Data Analysis:** Perform exploratory data analysis to identify trends and patterns.
- **Quantitative Risk Analytics:** Calculate important financial performance metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, VaR, and CVaR.
- **Behavioral Modeling:** Analyze investor behavior through cohort analysis and SIP continuity analysis.
- **Business Intelligence:** Build an interactive Power BI dashboard for data visualization and reporting.
- **Decision Support:** Generate meaningful insights that can support investment decision-making.

3 Technology Stack

The project uses a combination of programming languages, databases, visualization tools, and version control systems to complete the entire analytics workflow.

4 Dataset Description

This project uses multiple datasets related to the Indian mutual fund industry. Each dataset represents a different aspect of mutual fund operations, including fund details,

Technology	Purpose
Python	Data cleaning, analysis, and automation
Pandas	Data manipulation and preprocessing
NumPy	Numerical and mathematical calculations
Matplotlib	Statistical charts and visualizations
Plotly	Interactive visualizations
SQLite	Relational database for storing processed data
SQLAlchemy	Connecting Python with SQLite database
Jupyter Notebook	Data analysis and documentation
Power BI	Interactive dashboard development
Git & GitHub	Version control and project management
VS Code	Development environment

Table 1: Project Technology Stack

NAV history, investor transactions, portfolio holdings, SIP inflows, and benchmark indices. These datasets were cleaned, validated, and integrated before performing further analysis.

4.1 Fund Master

This dataset contains the master information for all mutual fund schemes, including AMFI code, fund house, scheme name, category, expense ratio, fund manager, launch date, and risk category.

Source File: 01_fund_master.csv

4.2 NAV History

This dataset stores the historical Net Asset Value (NAV) of all mutual fund schemes from 2022 to 2026. It is used for return calculations, trend analysis, and risk measurement.

Source File: 02_nav_history.csv

4.3 AUM by Fund House

This dataset contains the Assets Under Management (AUM) for different fund houses across multiple years. It is useful for comparing the market share and growth of asset management companies.

Source File: 03_aum_by_fund_house.csv

4.4 Monthly SIP Inflows

This dataset records monthly Systematic Investment Plan (SIP) inflows, active SIP accounts, and yearly growth percentages. It helps analyze investor participation and SIP growth trends.

Source File: 04_monthly_sip_inflows.csv

4.5 Category-wise Inflows

This dataset provides monthly net inflows for different mutual fund categories, helping identify investment preferences across categories.

Source File: 05_category_inflows.csv

4.6 Industry Folio Count

This dataset contains the total number of investor folios across equity, debt, hybrid, and other mutual fund categories.

Source File: 06_industry_folio_count.csv

4.7 Scheme Performance

This dataset includes important performance indicators for each mutual fund scheme, such as returns, Alpha, Beta, Sharpe Ratio, Sortino Ratio, expense ratio, and assets under management.

Source File: 07_scheme_performance.csv

4.8 Investor Transactions

This dataset stores investor transaction records, including transaction type, investment amount, state, city, age group, payment mode, and KYC status. It is used for investor analytics and cohort analysis.

Source File: 08_investor_transactions.csv

4.9 Portfolio Holdings

This dataset contains portfolio composition details, including stock names, sectors, allocation weights, market values, and portfolio dates. It is used for sector allocation and HHI concentration analysis.

Source File: 09_portfolio_holdings.csv

4.10 Benchmark Indices

This dataset includes historical benchmark index values used to compare mutual fund performance and calculate Alpha, Beta, and tracking error.

Source File: `10_benchmark_indices.csv`

5 Data Cleaning and ETL Process

The first step of the project was to prepare the raw datasets for analysis. The data was collected from multiple CSV files containing information about mutual funds, NAV history, investor transactions, portfolio holdings, SIP inflows, and benchmark indices.

During the data cleaning process, missing values, duplicate records, incorrect data types, and inconsistent values were identified and corrected. Date columns were converted into a standard date format, numerical columns were validated, and transaction categories were standardized to maintain consistency across the datasets.

After cleaning, all processed datasets were stored in the `data/processed` folder. The cleaned data was then loaded into a SQLite database using SQLAlchemy, creating a centralized database for further analysis and dashboard development.

5.1 Main ETL Steps

The extraction, transformation, and loading workflow followed a systematic pipeline:

1. Loaded all raw CSV files using the Pandas library.
2. Removed duplicate records across transactional and historical logs.
3. Converted date columns into a standard `datetime` format.
4. Validated numerical values such as Net Asset Value (NAV) and transaction amounts.
5. Standardized categorical text values for transaction types and KYC status.
6. Cleaned scheme performance variables and parsed expense ratio attributes.
7. Exported all cleaned datasets into the local processed data repository.
8. Loaded the finalized datasets directly into the SQLite database.

6 Database Design (SQLite)

A SQLite database was created to store the cleaned mutual fund datasets in a structured format. A ****Star Schema**** architectural pattern was chosen to organize the tables, decoupling static attributes from dynamic, time-series data. This optimization directly simplifies analytical queries and improves report rendering speeds.

The relational design cleanly separates data into structural dimensions and analytical fact tables. Relational integrity is enforced using primary keys and foreign keys, primarily tracking relationships through the standardized **AMFI Code** and **Date** fields.

6.1 Database Tables Schema

The relational architecture is segmented into the following modules:

6.1.1 Dimension Tables

- **dim_fund:** Contains core data defining individual assets, including scheme names, management metrics, and broad fund house classifications.
- **dim_date:** Provides time-intelligence variables to group data across days, months, quarters, and fiscal cycles.

6.1.2 Fact Tables

- **fact_nav:** Tracks transactional historical Net Asset Values on a daily frequency.
- **fact_transactions:** Captures atomic level user movements, values, payment choices, and demographic indexes.
- **fact_performance:** Aggregates mathematical performance dimensions, keeping track of calculations like absolute ratios and alpha-beta configurations.
- **fact_aum:** Holds asset management sizes logged across operational institutions.

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Figure 1: Tables of SQLite Database Schema Layout

7 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the mutual fund industry and identify meaningful trends before developing predictive metrics and dashboards. Different visualizations were created using Matplotlib, Seaborn, and Plotly to analyze NAV movements, AUM growth, SIP inflows, investor demographics, portfolio allocation, and category-wise investment patterns. The analysis helped identify market growth, investor participation, fund performance trends, and portfolio diversification.

7.1 NAV Trend Analysis

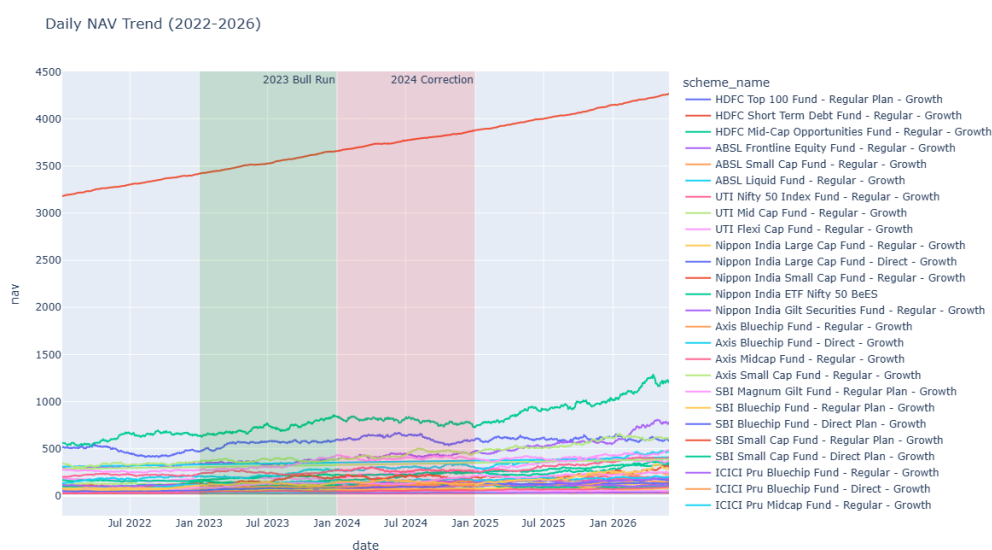


Figure 2: Historical NAV Trend Performance Analysis (2022–2025)

Key Findings:

- Most mutual fund schemes showed a steady increase in Net Asset Value (NAV) over the analysis period.
- Strong market growth was observed during 2023, indicating a bullish phase.
- A temporary decline was visible during the 2024 market correction.
- Long-term trends indicate consistent growth in equity-oriented schemes.

7.2 AUM Growth Analysis

Key Findings:

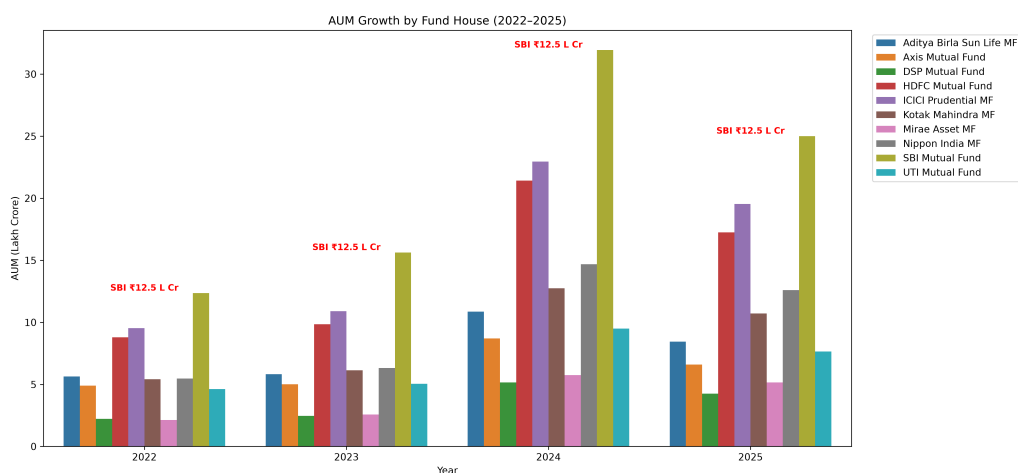


Figure 3: Assets Under Management (AUM) Growth Across Major Asset Management Companies

- SBI Mutual Fund maintained the highest Assets Under Management throughout the period.
- Most fund houses experienced consistent AUM growth between 2022 and 2025.
- Larger Asset Management Companies (AMCs) continued to dominate the mutual fund industry.
- Overall industry AUM increased significantly over the four years.

7.3 SIP Inflow Analysis

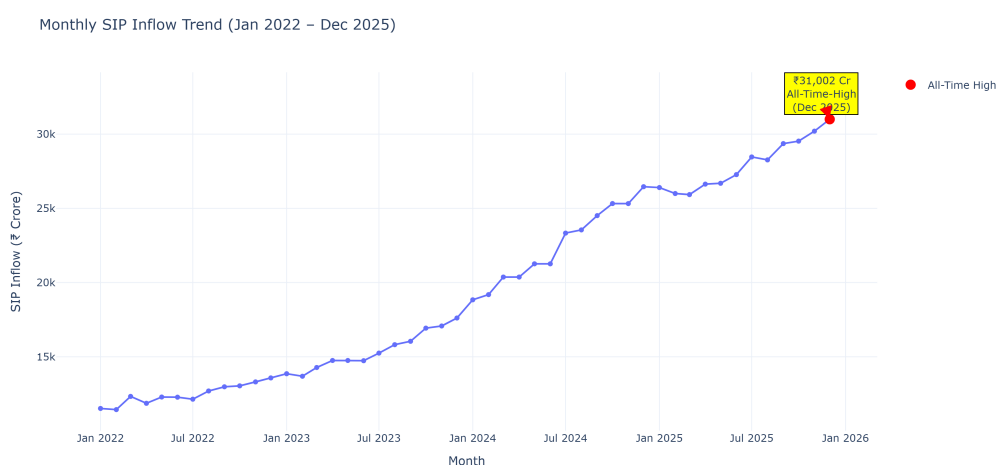


Figure 4: Monthly Systematic Investment Plan (SIP) Inflow Growth Vector

Key Findings:

- Monthly SIP collections increased steadily over time.
- The highest SIP inflow was recorded in December 2025 (Rs 31,002 Crore).
- Investor participation increased significantly after 2023.
- SIP continues to be one of the preferred investment methods.

7.4 Category-wise Inflow Analysis

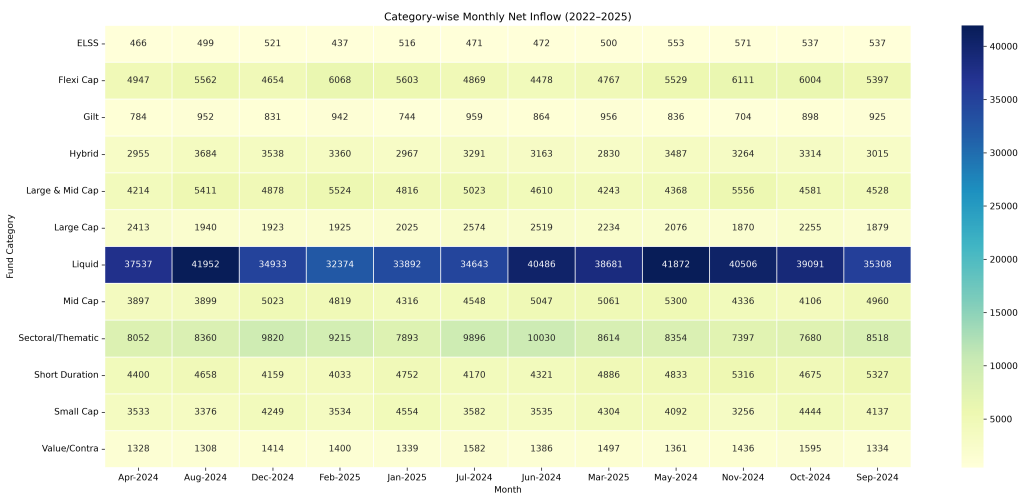


Figure 5: Heatmap Matrix of Net Asset Inflows Distributed by Fund Category

Key Findings:

- Equity categories received the highest net inflows.
- Debt fund inflows remained relatively stable.
- Hybrid funds showed moderate investor interest.
- Investment patterns varied across different months.

7.5 Investor Demographics

Key Findings:

- Investors aged 26–35 years contributed the highest number of investments.
- Male investors represented a larger share of total investors.
- Middle-aged investors generally invested higher SIP amounts.
- Young investors showed increasing participation in SIP investments.

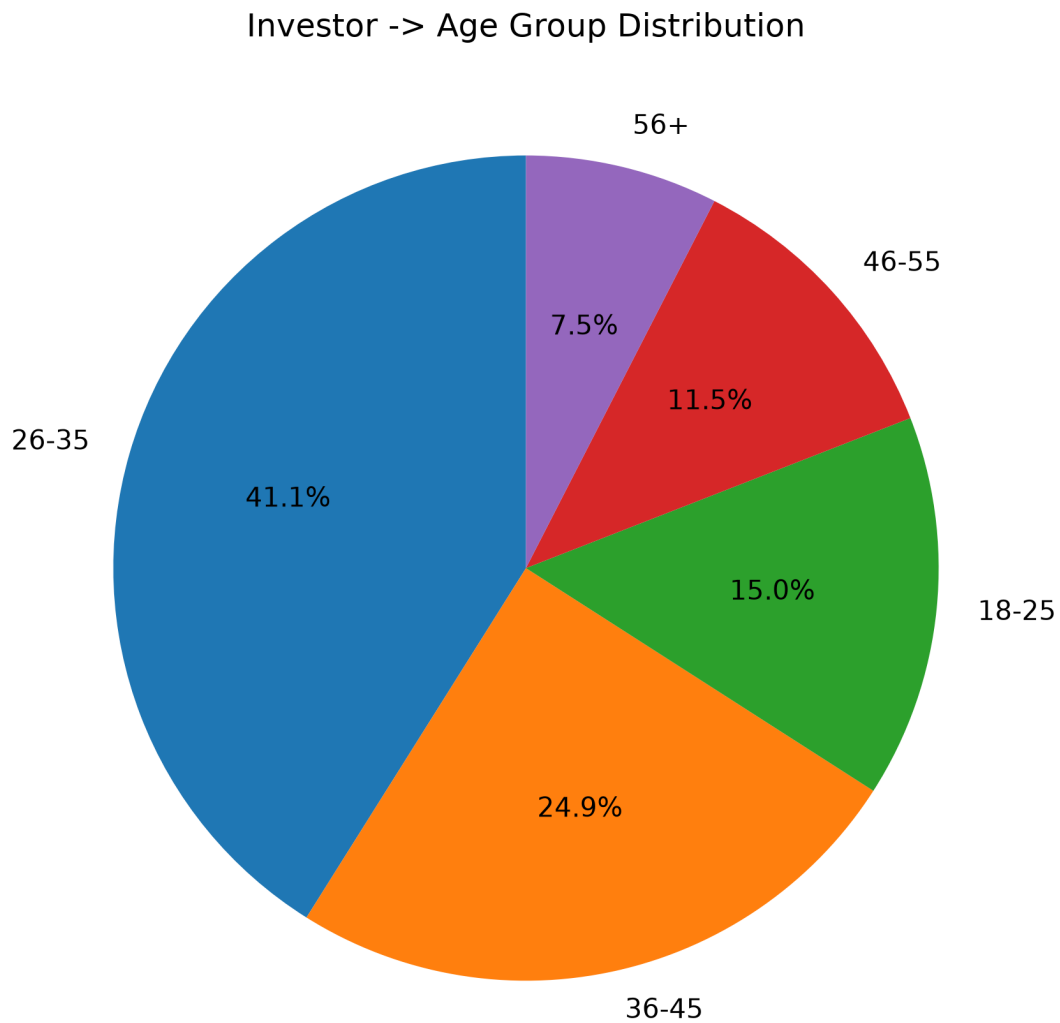


Figure 6: Investor Profile Metrics: Age Group Distribution, Gender Split, and SIP Matrix

7.6 Geographic Distribution

Key Findings:

- Metropolitan states contributed the highest transaction amounts.
- T30 (Top 30) cities accounted for a larger share of total investments.
- B30 (Beyond 30) cities showed steady growth in investor participation.
- Geographic diversification of investors continues to improve.

7.7 Sector Allocation

Key Findings:

T30 vs B30 - City Distribution

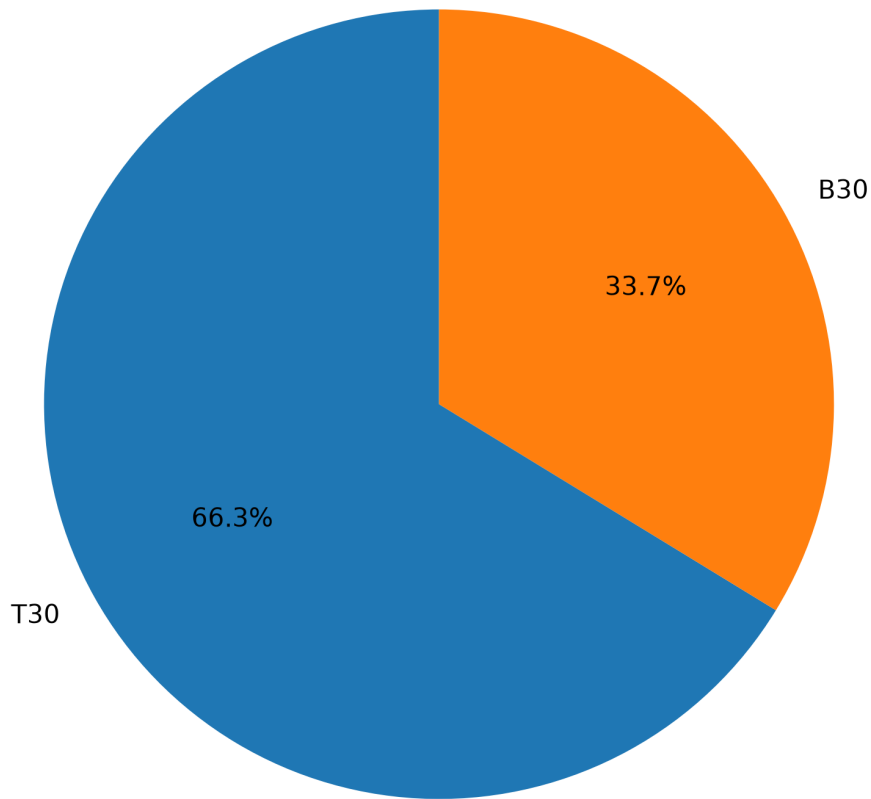


Figure 7: Geographic Allocation Mapping: State Contribution and T30 vs B30 Layout

- Financial Services formed the largest portfolio allocation.
- Information Technology and Healthcare were also major sectors.
- Portfolio allocation was well diversified across multiple sectors.
- No single sector dominated the entire portfolio allocation.

8 Performance Analytics

Performance analytics was performed to evaluate both the returns and risk associated with different mutual fund schemes. Several financial metrics were calculated using historical NAV data and benchmark indices. These metrics help compare funds based on profitability, consistency, and risk-adjusted performance.

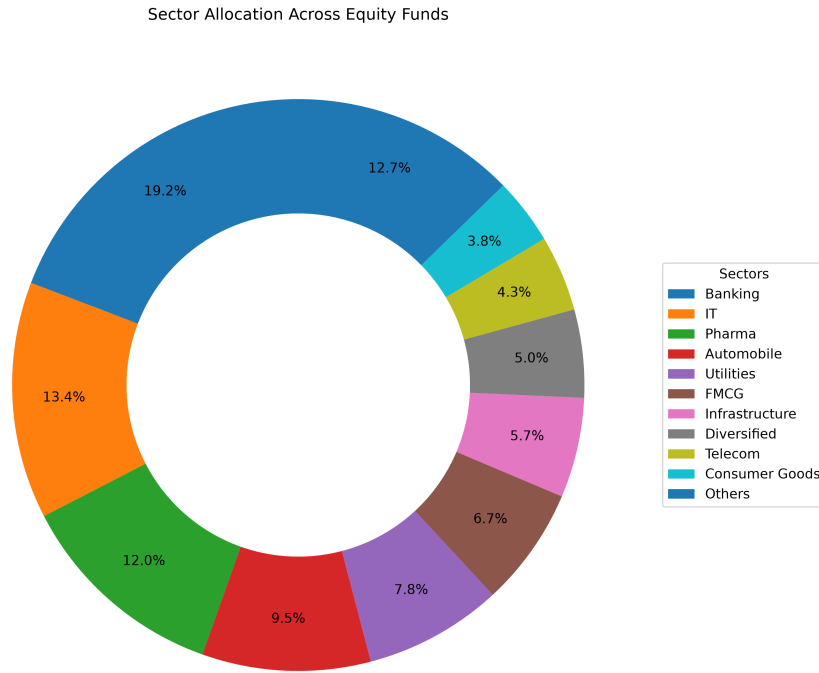


Figure 8: Portfolio Sector Concentration and Holdings Allocation Distribution

Key Findings:

- Most equity funds generated positive long-term returns.
- Three-year CAGR was generally higher than one-year CAGR.
- Consistent performers maintained strong returns across all periods.
- Long-term investments provided better wealth creation.

8.1 Sharpe Ratio

Key Findings:

- Higher Sharpe Ratio indicates better risk-adjusted returns.
- Top-ranked funds generated higher returns with controlled risk.
- Several funds outperformed their peers on a risk-adjusted basis.
- Investors seeking stable performance may prefer funds with higher Sharpe Ratios.

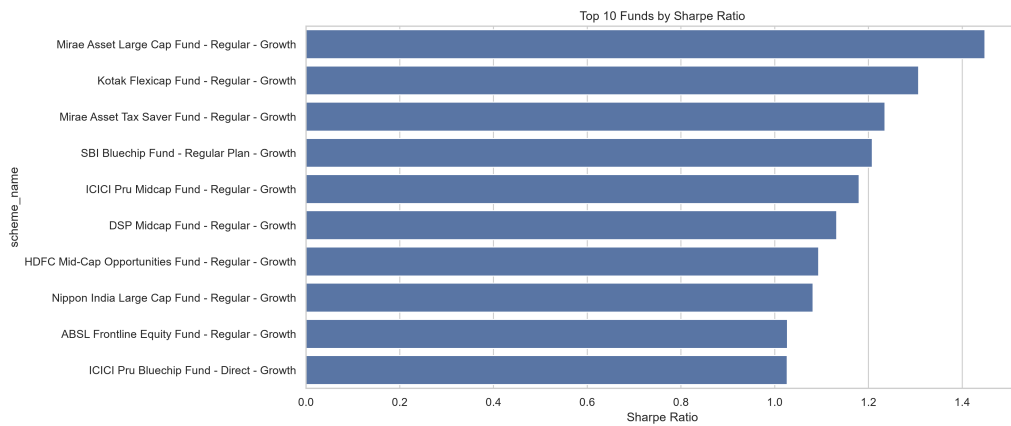


Figure 9: Risk-Adjusted Return Comparison Using Sharpe Ratio Rankings

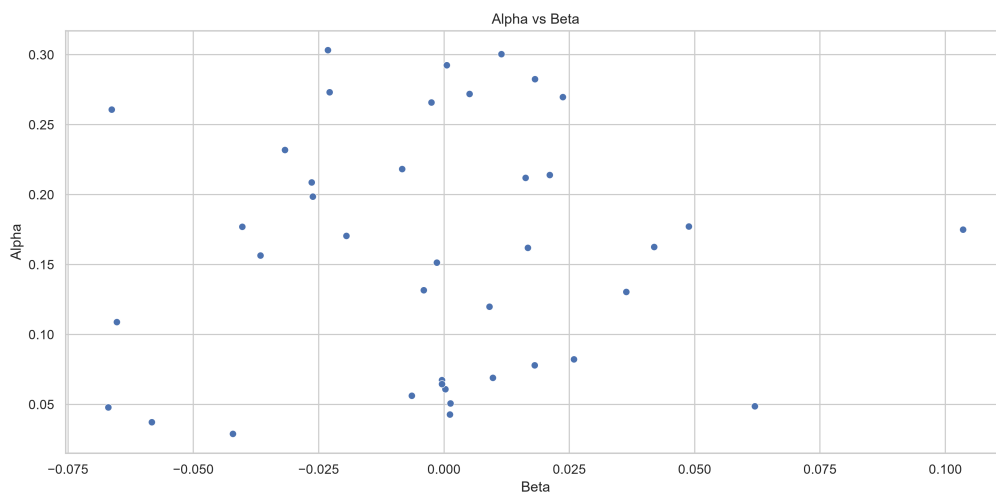


Figure 10: Volatilities and Excess Performance (Alpha vs. Beta Matrix)

8.2 Alpha and Beta Analysis

Key Findings:

- Positive Alpha indicates that a fund outperformed its benchmark index.
- Beta values close to one suggest market-like volatility structures.
- Higher Beta funds were more sensitive to broader market movements.
- Some funds consistently delivered positive excess returns.

8.3 Maximum Drawdown

Key Findings:

- Maximum Drawdown measures the largest peak-to-trough decline from a fund's value.

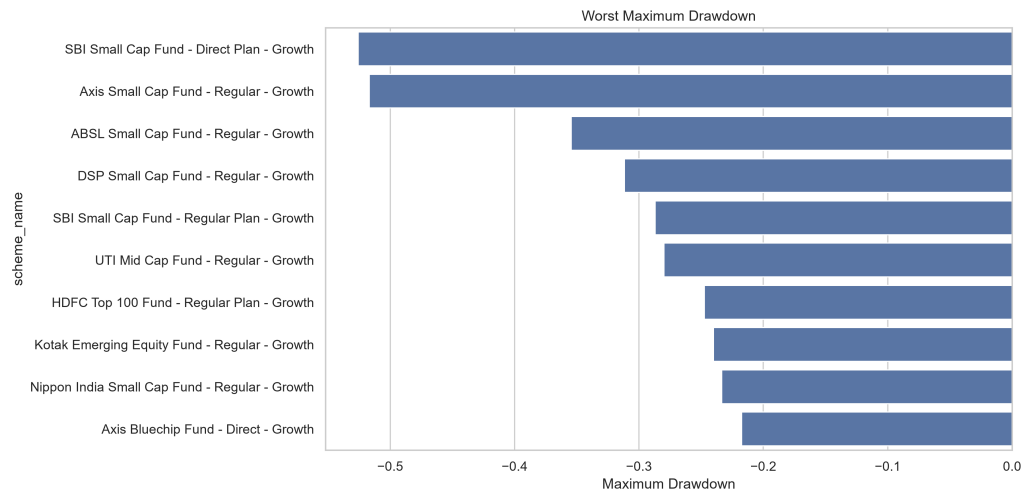


Figure 11: Downside Risk Assessment via Historical Maximum Drawdown Horizons

- Funds with lower drawdowns demonstrated superior downside protection capabilities.
- Market corrections affected some schemes significantly more than others.
- Investors generally prefer funds with structurally lower drawdown metrics.

8.4 Fund Scorecard

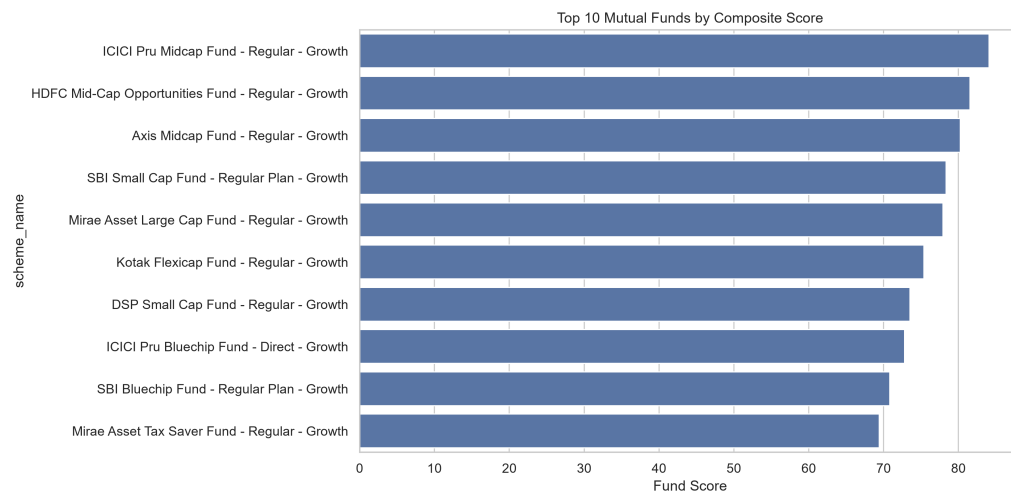


Figure 12: Comprehensive Consolidated Mutual Fund Performance Scorecard

Key Findings:

- The scorecard combines multiple performance metrics into a single weighted ranking system.
- Top-performing funds maintained strong returns alongside controlled volatility risks.

- Expense ratios explicitly exerted an inverse influence on the ultimate cumulative score.
- The scorecard matrix significantly simplifies relative mutual fund selection comparison workflows.

8.5 Benchmark Comparison



Figure 13: Relative Scheme Trajectory Tracking Against Benchmark Indices

Key Findings:

- Several funds consistently generated positive alpha over active benchmark indices.
- Benchmark comparison highlighted the absolute operational consistency of top-performing schemes.
- Tracking error remained optimally low for most large-cap funds.
- Active fund management effectively added visible value over pure passive benchmark performance.

9 Power BI Dashboard

An interactive Power BI dashboard was developed to present the analyzed mutual fund data in a simple and user-friendly format. The dashboard consists of four pages covering industry overview, fund performance, investor analytics, and SIP & market trends. Interactive slicers, drill-through functionality, and tooltips were added to improve user experience and make data exploration easier.

9.1 Dashboard Pages Architecture

- **Page 1 – Industry Overview:** Displays overall industry performance through operational KPI cards, AUM trajectories, and fund house market-share comparisons.
- **Page 2 – Fund Performance:** Provides relative comparison matrices of mutual fund returns, risk parameters, and benchmarks.
- **Page 3 – Investor Analytics:** Segments investor profiles using clean demographic layers and regional layouts.
- **Page 4 – SIP & Market Trends:** Tracks Systematic Investment Plan (SIP) inflow growth rates and macro sector trends over time.

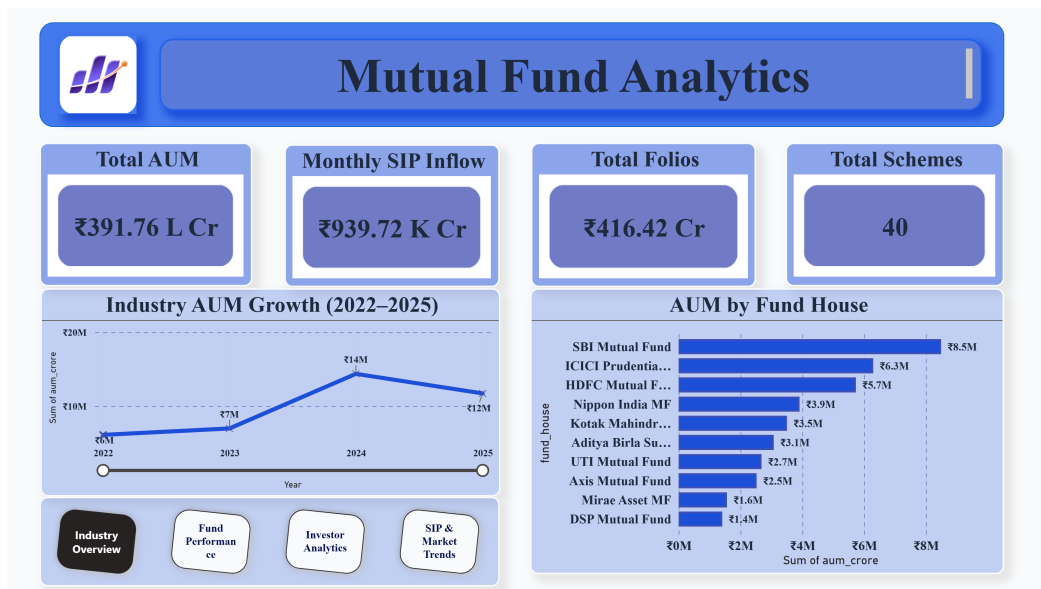


Figure 14: Power BI Dashboard Interface – Industry Performance & AUM Trends

9.1.1 Key Analytics Dashboard Insights

- The operational platform unifies disconnected distributed datasets to output a clear, single-source financial tracking cockpit.
- Parametric slicing engine allows fast filtering across diverse assets management institutions, investment schemes, and categorical classifications.
- Drill-through layers grant advanced analytical path routing directly into individualized mutual fund micro-level profiles.

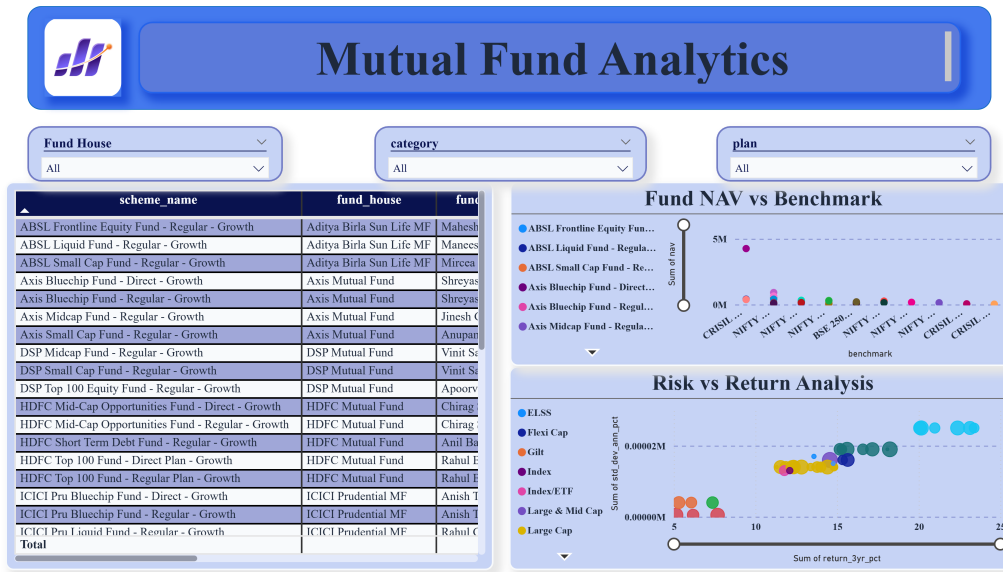


Figure 15: Power BI Dashboard Interface – Quantitative Scheme Risk-Return Benchmarking

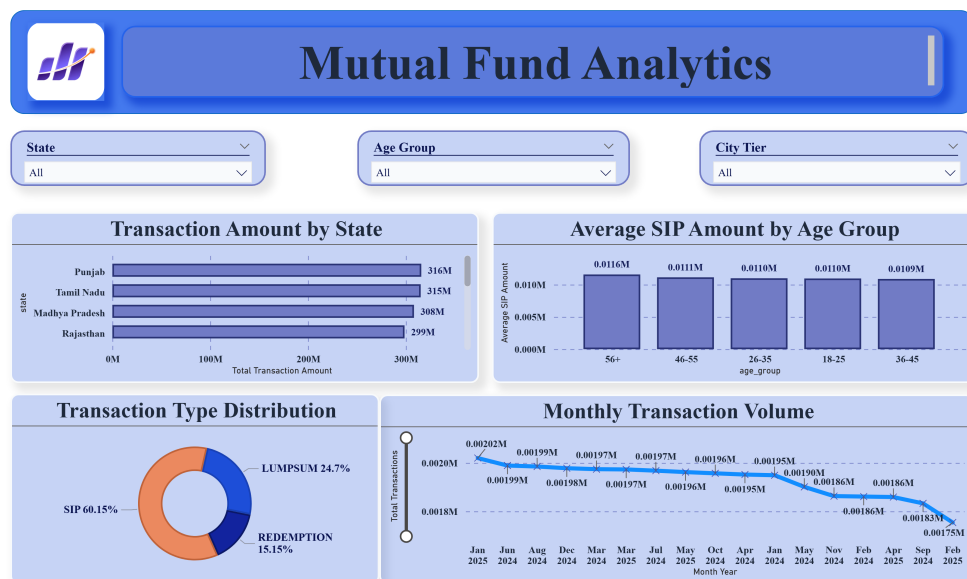


Figure 16: Power BI Dashboard Interface – Demographic Insights & Transaction Profiling

10 Advanced Analytics

Advanced analytical techniques were implemented to evaluate investment risk, investor behavior, and portfolio diversification. These analyses provide deeper insights beyond traditional financial metrics and help understand fund performance from different perspectives.

The analytical suite executes calculations covering Historical Value at Risk (VaR),

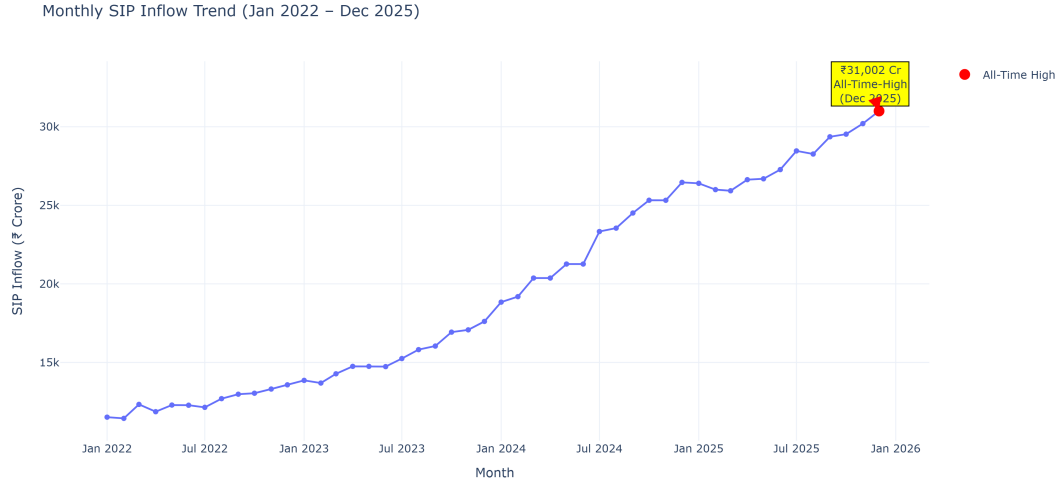


Figure 17: Power BI Dashboard Interface – Systematic Investment Inflows & Macro Sector Trends

Conditional Value at Risk (CVaR), Rolling Sharpe Ratio horizons, Investor Cohort Matrices, SIP Continuances, Portfolio Herfindahl-Hirschman Index (HHI) Concentrations, and a rule-based matching engine tailored to personalized investor risk tolerance categories.

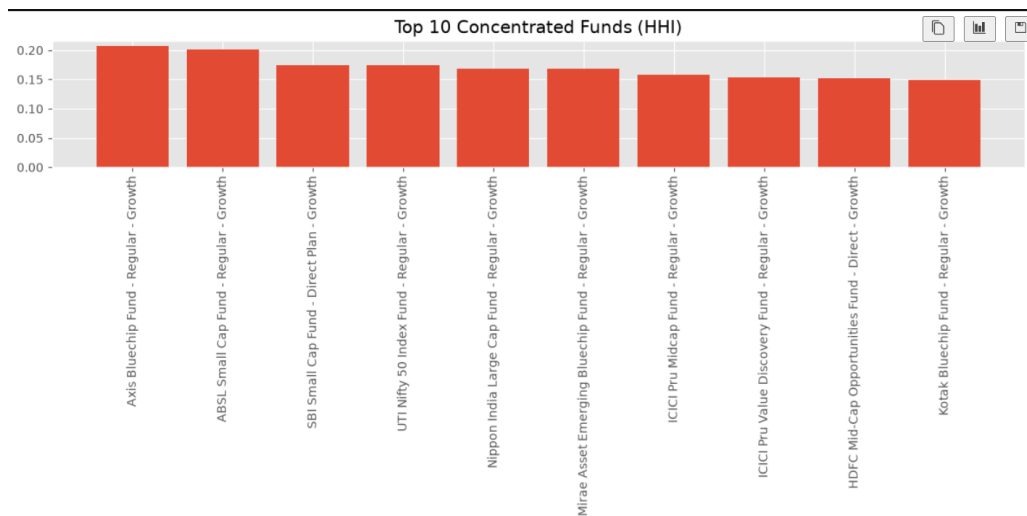


Figure 18: Herfindahl-Hirschman Index (HHI) Industry Sector Exposure Dispersion

10.0.1 Key Structural Discoveries

- Tail-risk configurations identified high-volatility thematic funds having a significantly deeper downside tail profile, requiring sound buffer adjustments during sudden economic drawdowns.
- Time-sequenced investor cohorts launched across recent operational windows demon-

strate higher relative starting capitalization sizes compared to historically mature cohorts.

- Systemic calculation of portfolio asset distribution weights using the HHI algorithm safely flagged highly concentrated equity funds matching structural over-exposures to single target industries.

11 Key Findings

The extensive analysis performed across data engineering pipelines, mathematical modeling engines, and transactional operational data surfaced several critical conclusions:

- **Long-Term Resilience:** Fund Net Asset Values highlighted structurally continuous historic multi-year up-trends despite localized macro-market corrections.
- **Asset Supremacy:** SBI Mutual Fund successfully maintained its absolute lead holding the highest volume market share across Assets Under Management metrics.
- **Retail Investment Surge:** Monthly retail capital inflows across Systematic Investment Plans recorded an all-time peak high value of Rs 31,002 Crore during December 2025, validating robust retail confidence frameworks.
- **Demographic Focus Point:** Younger retail demographics matching the 26–35 age range represented the core transaction generation pillar for automated monthly investments.
- **Risk-Adjusted Outperformance:** Schemes holding mathematically higher Sharpe and Sortino parameters demonstrated superior asset preservation traits alongside competitive returns.
- **Alpha Generation Profile:** A significant cluster of active equity-oriented funds generated positive excess returns (Alpha) relative to their tracking baseline index portfolios.
- **End-to-End System Efficiency:** Merging Python script processing steps, low-latency SQLite transactional database schemas, and clean Power BI dashboard front-ends generates a scalable, cost-efficient, industrial-grade data analytic stack.

12 Conclusion

This project successfully developed an institutional-grade, end-to-end mutual fund analytics solution by integrating data engineering, advanced quantitative financial analysis, and business intelligence techniques. Scalable Python pipelines effectively parsed and structured raw chaotic data, storing normalized records into a performant SQLite database layout.

Rigorously evaluating performance stability using metrics spanning traditional risk-reward formulas alongside extreme downside tail metrics (such as historical Value at Risk) delivers a definitive model for classifying portfolio performance. Ultimately, the interactive Power BI dashboard framework succeeds in translating dense, confusing financial rows into simple, action-oriented graphical visualizations. This project proves how modern data analytics tools can turn raw numbers into valuable insights, enabling smarter investment strategies and a better understanding of the mutual fund market.

13 Future Scope

While the current framework securely achieves its core operational targets, the architecture can be scaled forward via several strategic enhancements:

1. **Real-Time Pipelines:** Connect cloud-hosted programmatic ingestion endpoints to feed active real-time NAV ticks straight into the database engine.
2. **Predictive Modeling:** Introduce machine learning architectures (such as LSTM networks or gradient boosting trees) to extract trend predictions out of historical time-series data.
3. **Portfolio Optimization Engine:** Implement a mathematical asset-allocation framework based on Modern Portfolio Theory (MPT) to dynamically suggest optimal weights.
4. **Native Web Deployment:** Port dashboard elements into modern web frameworks (such as Streamlit or custom React containers) to increase overall application accessibility.
5. **AI Advisory Layer:** Deploy natural language processing modules to generate automated narrative interpretations of portfolio scorecards tailored to custom risk profiles.